

PICOPATT Internship

Checkpoint + 1-Month Plan — May 2026

Explainable Clustering for Spatio-Temporal Picoclimatic Data

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Snapshot



- All numbers and figures are from archived outputs and notebook runs.

Standard techniques

- Finite differences (Δ , Δ^2 , Δ^3)
- Multi-scale shapelets in literature
- Jaccard overlap for attribution stability

Defensible contributions

- Stability-optimized mixed-length dictionary selection
- Sensor-unit explanation format as a fixed contract
- Four-term objective with spatial + temporal penalties
- Alignment-gap taxonomy for planner-facing XAI

Core Contribution Framing

- Application-driven method to align representation, clustering, explanation.
- Explanations must remain in physical units across data granularities.
- Stability is reported explicitly, not inferred.

Joint objective (conceptual)

$$L = L_{km} + L_{sh} + L_{tc} + L_{stab}$$

KMeans fit + spatial homogeneity + temporal coherence + stability penalty.

1-Month Plan (May 21 – June 20)

- Week 1: impute raw slots, rerun $\Delta/\Delta^2/\Delta^3$ baseline.
- Week 2: map feature indices to physical sensor units in explanations.
- Week 3: expand stability to 10 seeds and finalize tables.
- Week 4: manuscript polish; optional joint-objective prototype.

Immediate Decision

Option A

- Fix derivative baseline with imputation
- Re-run clustering metrics and stability

Option B

- Sensor-unit mapping for explanations
- Freeze the explanation contract

Risks and Mitigations

- Risk: derivatives degrade clustering after imputation.
- Mitigation: report negative result; favor statistical summaries.
- Risk: 10-seed sweeps are slow on Windows + MKL.
- Mitigation: keep bounded sampling and log runtime.